

Design of Spur-Free $\Sigma\Delta$ Frequency Tuning Interface for Digitally Controlled Oscillators

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Abstract—Within the framework of an RF all-digital phase-locked loop (ADPLL), frequency tuning of a digitally controlled oscillator (DCO) is realized using switchable capacitors (varactors). The minimum feature size and lithographic mismatch of these varactors limit the physical frequency resolution of the DCO to 10's of kHz. To satisfy the fine frequency resolution of 100's of Hz in wireless cellular and connectivity standards, a high-speed $\Sigma\Delta$ frequency dithering is conventionally employed to improve the time-averaged DCO resolution by at least two orders of magnitude. Unfortunately, the dithering characteristics of the $\Sigma\Delta$ modulator vary depending on its fractional frequency input and the physical and timing mismatches in the mapping of the multi-bit digital output to frequency. This paper investigates dithering mechanisms that are a potential source of spurious corruption of the DCO output spectrum and proposes a set of enhancements to mitigate them. Furthermore, the spurious sensitivity to design parameters, such as device mismatch as well as rise/fall, duty-cycle and skew timing mismatches, is theoretically analyzed and verified by behavioral simulations.

Index Terms—Phase-locked loop, digitally controlled oscillator, frequency synthesis, frequency dithering, sigma-delta modulator.

I. INTRODUCTION

WITH the recent advancements of nanometer-scale CMOS technology, all-digital phase-locked loops (ADPLLs) have gained widespread interest due to their flexibility, reconfigurability, transfer function stability, settling speed, precise frequency modulation capability and amenability to integration with digital processors [1] [2] [3] [4]. The centerpiece in an ADPLL is a fine-resolution digitally controlled oscillator (DCO) that comprises a resonating LC tank and quantized frequency tuning units realized with digitally switched capacitors (varactors) [5].

Fig. 1 shows a block diagram of the fine frequency control of a multi-GHz DCO. The DCO tuning control is split into several banks of varying degree of frequency step size and range: coarse d^P for process, voltage and temperature (PVT) calibration of the frequency band of operation; medium d^A for channel acquisition; fine d^{TI} and d^{TF} for fine frequency tracking of the oscillator drift and modulation. The capacitor

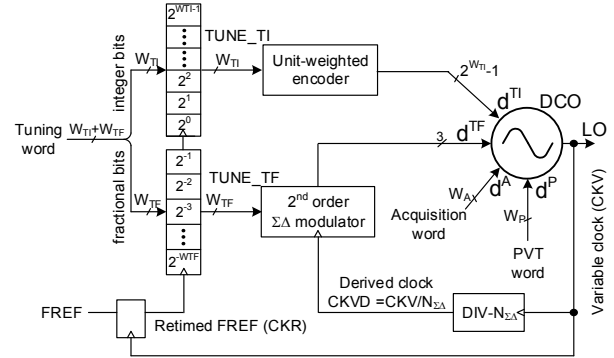


Fig. 1. Frequency control of a DCO core using multiple capacitor banks.

banks are built using Metal-Insulator-Metal (MIM) and/or Metal Oxide Semiconductor (MOS) varactors [1] [5].

The finest varactor step size feasible in the advanced CMOS lithography is on the order of 10's of attofarads (aF). For example, a 40 aF capacitance step corresponds to a 12 kHz frequency step at the 2 GHz DCO output. Unfortunately, this fine capacitance step is still too coarse for wireless standards, so dithering is used to improve the *time-averaged* capacitance resolution [1]. A typical realization, which balances the need for adequate dithering randomization while not violating the far-out DCO phase noise spectrum, uses a second-order MASH $\Sigma\Delta$ modulator (SDM) [1] with 8 fractional input bits. Such an arrangement will produce a sufficiently-fine open-loop frequency resolution of $12 \text{ kHz}/2^8 = 47 \text{ Hz}$. However, a textbook implementation of such a dithering scheme may result in potentially ill-behaved spurs in the DCO output spectrum, as discussed in Section II. These spurs are further degraded by varactor and timing mismatches.

Extensive studies [6] [7] [8] [9] [10] have attempted to address the SDM spurs, mainly in the field of digital-to-analog conversion (DAC), where the high frequency noise/spurs are easily filtered out by a reasonably strong low-pass filter. In [2], the DCO is realized as a $\Sigma\Delta$ DAC and a VCO. This allows an analog low-pass filter to be inserted after the SDM, thus exploiting standard techniques developed for $\Sigma\Delta$ data converters. However, to capitalize on the smaller silicon footprint, lower consumed power, better noise immunity and wider tuning bandwidth, the SDM output has to control the oscillator core digitally without any intervening analog filtering. Despite sharing some common features, the design constraints for a DCO $\Sigma\Delta$ interface are quite different from those in a $\Sigma\Delta$ DAC, hence the motivation for this research work. The following sections focus on the DCO $\Sigma\Delta$ dithering

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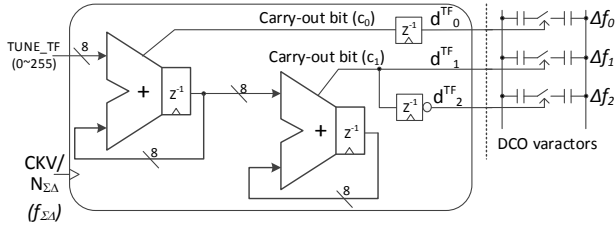


Fig. 2. 2nd-order MASH $\Sigma\Delta$ modulator used for fractional frequency dithering of DCO varactors.

scheme, while investigating mechanisms causing undesired spurs followed by proposed methods to suppress them.

II. $\Sigma\Delta$ DITHERING SCHEME AND SPURS

Fig. 2 shows a digital 2nd-order MASH $\Sigma\Delta$ modulator [1] [5], which converts an $W_{TF}=8$ -bit binary fractional part, $TUNE_TF$, of the tuning code of weight $1/2^8$ to a high-rate 3-bit unit-weighted code ($d_{0,1,2}^{TF}$) of weight 1 that directly controls three digital varactor units. Mathematically, the output of the $\Sigma\Delta$ modulator is expressed as:

$$out^{\Sigma\Delta}[k] = d_0^{TF} + d_1^{TF} + d_2^{TF} = c_0[k-1] + c_1[k] - c_1[k-1] \quad (1)$$

where, k is the discrete-time index, c_0 and c_1 are carry-out bits of the two accumulators. The summation of (1) is physically realized using a parallel arrangement of digital varactors. The instantaneous value of the fractional frequency tuning f^{TF} (in Hz units) executed by the DCO is expressed as,

$$f^{TF}[k] = c_0[k-1]\Delta f_0^T + c_1[k]\Delta f_1^T - c_1[k-1]\Delta f_2^T \quad (2)$$

where Δf_0^T , Δf_1^T and Δf_2^T are frequency steps (in Hz units) of the three digital varactors. Ideally, $\Delta f_{0,1,2}^T = \Delta f^T$, so $f^{TF}[k] = out^{\Sigma\Delta}[k] \cdot \Delta f^T$ and the time averaged DCO frequency equals to the fractional frequency command $TUNE_TF/2^8 \cdot \Delta f^T$, thus an equivalent frequency resolution of $\Delta f^T/256$ is achieved.

The noise transfer function of 2nd-order MASH $\Sigma\Delta$ modulator in Fig. 2 is expressed in z-domain as [11]:

$$NTF(z) = (1 - z^{-1})^2 \quad (3)$$

where $\Sigma\Delta$ modulator is running at a sampling clock $CKV/N_{\Sigma\Delta}$ of frequency $f_{\Sigma\Delta}$. The frequency-domain dithering noise power ($N_{\Sigma\Delta}$) can be derived as $|N_{\Sigma\Delta}(f)|^2 = |1 - \cos(\pi f/f_{\Sigma\Delta})|^2$

The phase noise resulting from the $\Sigma\Delta$ frequency dithering for a typically fractional input at a frequency offset f is given by [5]:

$$PN(f) = \frac{(\Delta f^T)^2}{12 \times f_{\Sigma\Delta}} \times |N_{\Sigma\Delta}(f)|^2 \times \frac{1}{f^2} \quad (4)$$

For most applications, Δf^T and $f_{\Sigma\Delta}$ are chosen such that, at any frequency, the dithering noise energy is well below the oscillator intrinsic thermal noise. This way, the noise impact of the DCO dithering would be negligible. Fig. 3 shows the $\Sigma\Delta$ dithering phase noise, computed from (4), for a nominal 2 GHz DCO, with a Δf^T of 20 kHz and a 2nd order $\Sigma\Delta$ modulator clocked at 500 MHz (or 1/4th of the DCO frequency). The dashed curve shows the targeted intrinsic phase noise profile of a wireless cellular standard-compliant oscillator.

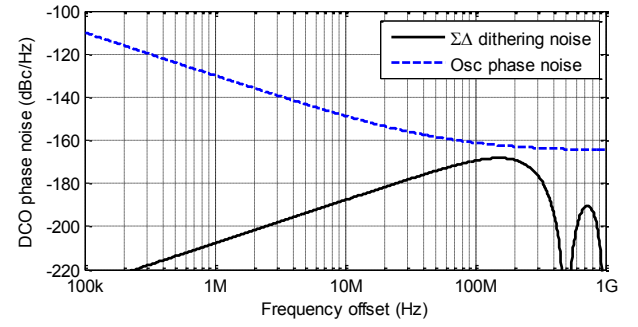


Fig. 3. Phase noise due to an ideal $\Sigma\Delta$ modulator versus intrinsic phase noise of a typical cellular mobile DCO.

It is well known that a $\Sigma\Delta$ output may embody dominant frequency modes for certain fractional inputs [11]. Furthermore, there are several physical impairments in the dithering implementation that can cause a dithering noise profile to deviate from (4) and may result in spurs at the DCO output. The key sources of spurs are described in the following subsections followed by one proposed scheme to completely suppress all spurs.

A. $\Sigma\Delta$ Modulator Idle Tones

A key assumption in (4) is that the $\Sigma\Delta$ stimulus has a uniform distribution in both frequency and time domains. While this assumption generally holds true for $\Sigma\Delta$ modulators in analog mixed-signal data converters [11], where an analog input signal spans multiple quantization levels, this is hardly the case for the DCO input when a low-noise ADPLL is locked to a fixed channel frequency. In polar modulators, this is circumvented by the frequency modulation signal that is added to the DCO control word [4].

With a constant input to the $\Sigma\Delta$ modulator shown in Fig. 2, the accumulated values in both accumulators, as well as 3-bit $\Sigma\Delta$ modulator output may repeat at N_{idle} modulator clock cycles, and can potentially result in idle tones at $f_{idle} = f_{\Sigma\Delta}/N_{idle}$ and its harmonic frequencies [12]. Generally, with an input of 2^M , the output of a $\Sigma\Delta$ modulator with N -bit accumulators ($N \equiv W_{TF}$ in Fig. 1) repeats every 2^{N+1-M} clock cycles. Consequently, the fundamental idle tone frequency (or the minimum idle tone frequency) of a $\Sigma\Delta$ modulator with an arbitrary input can be expressed as:

$$f_{idle}(N, M) = \frac{f_{\Sigma\Delta}}{2^{N+1-M}} \quad (5)$$

where M is a number of consecutive zeros at LSBs of the input word.

B. Mismatch of Varactor Frequency Steps

The three $\Sigma\Delta$ modulator output bits of Fig. 2 are mapped to three separate fine frequency tuning units. If the frequency steps of these three units are not identical (i.e., Δf_0^T , Δf_1^T , Δf_2^T are not equal to Δf^T), the implementation is no longer equivalent to (3) and (4). For example, if Δf_0^T is different from Δf_1^T and Δf_2^T , i.e., $\Delta f_1^T = \Delta f_2^T = \Delta f^T$ and $\Delta f_0^T = \Delta f^T + \delta f^T$, then the fractional tuning frequency can be written as:

$$\widehat{f^{TF}}[k] = (c_0[k-1] + c_1[k] - c_1[k-1])\Delta f^T + c_0[k-1]\delta f^T \quad (6)$$

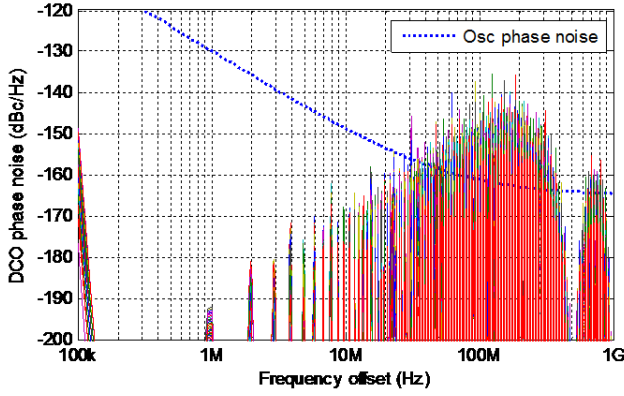


Fig. 4. DCO $\Sigma\Delta$ dithered spectrum for a set of constant fractional inputs in face of a $\pm 5\%$ frequency tuning bank varactor mismatch.

In short, the fractional dithered frequency $\widehat{f}^{TF}$ is equivalent to an ideal 2^{nd} -order $\Sigma\Delta$ modulator (SDM) producing f^{TF} , plus a first-order SDM driving a varactor of step size of $(\Delta f_0 - \Delta f^T)$. This additional term has less randomization than the ideal modulator and can potentially harbor accumulator modes that can be a possible source of observable spurs.

Fig. 4 shows the phase noise due to the $\Sigma\Delta$ DCO dithering of Fig. 2 for 8-bit static inputs of 1, 2...255. The frequency step sizes of the fractional tracking bank are 20 kHz with $\pm 5\%$ mismatch. It is observed that a large number of spurs can be generated that violate the noise level set by the oscillator intrinsic phase noise. These spurs cannot be characterized using a simple noise model shown in (4), and need to be carefully investigated to ensure that the overall phase noise and spurious emission specifications are met under all conditions.

C. Mismatch of Varactor Input Rise/Fall Times

The rising/falling mismatch (or duty cycle error) of the three SDM output bits results in a frequency error as well as an additional source of spurs. It is, therefore, desirable to shorten the physical routing from the digital SDM to the DCO frequency tuning units. It is best to avoid the need for a large number of buffer circuits or possible asymmetric rising/falling times and loading to avoid introduction of large rising and falling edge mismatches.

D. Mismatch of Varactor Input Delays

Since the SDM output is three bits wide and it needs to propagate all the way to the DCO tuning circuitry, it is likely that the individual bits will experience different arrival times due to the routing mismatches. Following the best design and layout practices should minimize this mismatch but it will not completely eliminate it. The impact of these bit transition mismatches is similar in behavior to the frequency step mismatch and could be another source of spurs resulting from the DCO varactor dithering.

As illustrated in Fig. 5 example, d_2^{TF} has a longer delay than the other two bits. There is no impact of this delay during DCO time periods of T_1 and T_3 . The DCO periods of T_2 and T_4 are impacted, i.e., the effective time-averaged frequency step is distorted to $d_2^{TF'}$ instead of the nominal contribution d_2^{TF} . This results in the distortion of the effective dither value applied to the DCO. Such a dynamic mismatch can also result in spurs at the DCO output spectrum.

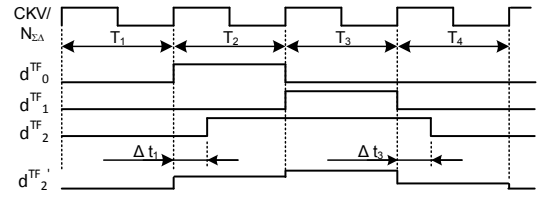


Fig. 5. Illustration of the DCO frequency tuning bit delay mismatch.

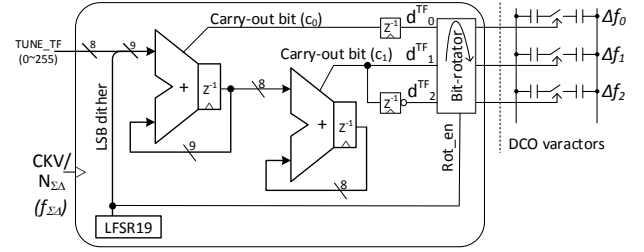


Fig. 6. Spur-free DCO tuning interface with two proposed techniques: LSB dithering of SDM and bit rotation of SDM output.

III. SPUR-FREE $\Sigma\Delta$ DITHERING SCHEME

To suppress the spurs arising from the DCO fine frequency tuning mechanism, some dithering and bit mapping schemes are necessary to remove all possible periodicities in the DCO control words. Simply mapping the $\Sigma\Delta$ output bits into a thermometer-coded word is inadequate to reduce the spurs. Fig. 6 shows the proposed DCO dithering scheme.

1) *Mitigation of Spurs due to Idle Tones*: A key observation from (5) is that the fundamental idle tone frequency switches between a lower and a higher idle frequency for an input numerator change of one. Consequently, if a single-bit random sequence is added to the input, the fundamental frequency may randomly switch between a higher and the minimum idle tone frequency, causing a distribution of the energy in idle tones in the frequency domain, thus eliminating most observable idle tones in the spectrum.

In a DCO, low frequency noise introduced by the dithering of LSB input is typically not a concern as the $\Sigma\Delta$ noise can be designed to be far below the intrinsic phase noise of the oscillator. At higher frequency offsets, the SDM noise contribution is more critical because of the stringent emission requirements of wireless transmitters (see Figs. 3 & 4). However, the LSB dithering noise is naturally filtered out by the inherent integration in the oscillator, that rolls off with a -20 dB/dec slope, as expressed below:

$$PN_{LSBdither}(f) = \frac{(\Delta f^T)^2}{2^{2N+2} \times f^2 \times f_{dither}} \quad (7)$$

where, f_{dither} is the data rate of the 1-bit pseudo-random sequence. Consequently, the total dithering noise is the sum of (4) and (7) in the absence of other impairments.

2) *Mitigation of Spurs due to Step Mismatch*: To eliminate the spurious tones caused by the frequency tuning unit mismatch, a random bit rotator, as shown in Fig. 6, is inserted between the SDM output and the varactors. In the proposed scheme, the 3-bit $\Sigma\Delta$ modulator output is dynamically mapped to three frequency tuning units. The bit mapping table is rotated in the sequence of $[2,1,0] \rightarrow [0,2,1] \rightarrow [1,0,2] \rightarrow [2,1,0]$... if the signal Rot_en is 1, otherwise, the bit mapping table

stays unchanged. The signal Rot_en is generated using an LFSR pseudo-random bit generator so as not to have any short-term periodicity and to reduce the concentration of spurious energy spectra. Note that for an efficient implementation, the random bit sequence used above for the LSB dithering to mitigate the idle tones can be shared to drive the bit rotator as well.

The bit rotator randomizes the energy caused by the tuning step mismatches, resulting in white noise at the DCO input, which is then integrated by the $1/s$ transfer function of the oscillator. The resulting phase noise is well-behaved:

$$PN_{MMdither}(f) = \frac{(\sigma f^T)^2}{2 \times f^2 \times f_{\Sigma\Delta}} \quad (8)$$

where, σf^T is the standard deviation of the tuning step sizes.

3) *Mitigation of Spurs due to Delay Mismatch:* As described in Section II, the impact of the delay mismatch of the frequency tuning bits has a similar impact as the tuning step mismatch, however its impact is dynamic and occurs only during the bit transitions. Let us assume the rising/falling delay mismatch of the tuning bits has a standard deviation of σt_{rf} . The duty-cycle errors result in an equivalent unit mismatch standard deviation of $\sigma t \times f_{\Sigma\Delta} \times \Delta f^T$ at the bit transitions and the noise energy after randomization by the bit rotator can be expressed as:

$$PN_{DCdither}(f) = \frac{(\sigma t_{rf} \times f_{\Sigma\Delta} \times \Delta f^T)^2}{2 \times f^2 \times f_{rotate}} \quad (9)$$

The above dithering scheme and (9) only deal with the standard deviation of the duty-cycle errors, and the bit rotator shown in Fig.6 does not spread out the noise energy for common duty-cycle errors on all tuning bits. This may result in introducing a static frequency error in the DCO. Simulation results show that the rising/falling delay mismatch should be no more than 20 ps. In case the average rising/falling delay mismatch among the $\Sigma\Delta$ output bits is beyond the desired tolerance, a modified mitigation scheme shown in Fig. 7, may be employed.

4) *Improved Bit Rotation Scheme for Delay and Rise/Fall Time Skew:* In this scheme, a dummy bit is combined with three $\Sigma\Delta$ modulator output bits and fed to a 4-bit random bit rotator, whose operation principle is similar to the 3-bit random bit rotator discussed before. Every other bit is inverted at the output of the bit rotator before it goes through a matched routing bus. In the frequency tuning bank of the DCO, every other bit is inverted again before it is connected to the corresponding tuning units. In this scheme, the average rising/falling delay mismatch is canceled for each consecutive pair of control bits, and the random bit rotator randomizes the rising/falling delay mismatch among the four control bits to minimize any aggregated spur tone energy.

The dithering scheme in Fig. 7 further eliminates the spurs caused by the delay mismatch among tuning units. Similarly to the rising/falling delay mismatch, the delay mismatch only has an impact at the data transitions. However, the delay mismatch results in less energy at lower frequencies because both the rising and falling edges of each tuning bit have the same amount of delay mismatch and hence cancel each

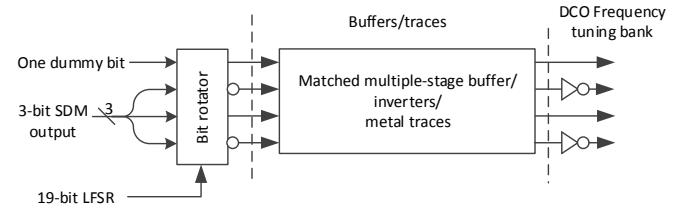


Fig. 7. Scheme to minimize the static rising/falling edge and delay mismatches.

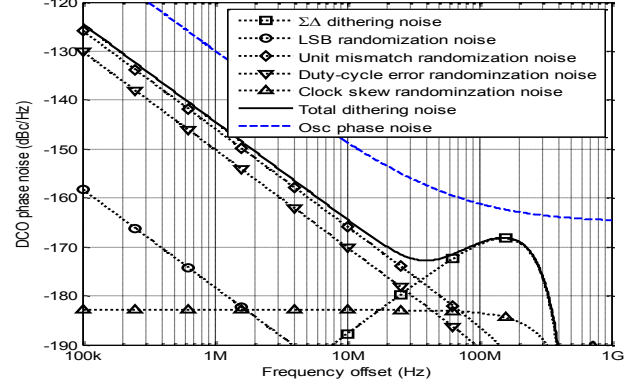


Fig. 8. Calculated noise profiles of the proposed $\Sigma\Delta$ dithering schemes in the presence of idle tones, $\pm 10\%$ varactor mismatch, 150 ps rise/fall time mismatch and 400 ps delay mismatch.

other, resulting in zero energy at dc. Therefore the impact of such a delay mismatch is more problematic only at higher frequencies. While this impact is negligible for $\Sigma\Delta$ data converters that are followed by a high-order low-pass filter, it is quite important for DCO applications due to only one pole filtering implicit in the DCO. This impairment can cause spurs at high frequency offsets in the absence of a spur mitigation.

With the randomization scheme shown in Figs. 6 & 7, the spurious energy is dithered out such that it exhibits a high-pass noise profile at the SDM output. Cascaded with the integration function of the oscillator, the noise finally appears as white noise in the DCO output spectrum, as expressed in (10), where σt_s is the standard deviation of the delays of $\Sigma\Delta$ output bits.

$$PN_{SKdither}(f) = \frac{4 \times (\sigma t_s \times f_{\Sigma\Delta} \times f^T)^2}{f_{\Sigma\Delta}^4} \quad (10)$$

Fig.8 shows an example of all noise profiles analyzed above, in which varactor mismatch is $\pm 5\%$, rising/falling mismatch is 150 ps, and the delay mismatch is 400 ps. As long as the total dithering noise (bold curve) is buried well below the oscillator intrinsic phase noise (dashed blue curve), the dithering noise will have no observable effect on the DCO output spectrum. With the analysis presented above, one can easily determine the specifications for the device mismatch, transition time and delay mismatches based on the targeted phase noise profile in an application.

IV. SIMULATIONS

Simulations for all possible input values (1..255) of the 8-bit fractional frequency tuning word, TUNE_TF, are carried out to validate the analysis in the previous sections as well as to show effectiveness of the proposed low-complexity spur-free DCO interference. For all simulations here, the nominal frequency step size f^T of the varactors is 20 kHz, and the

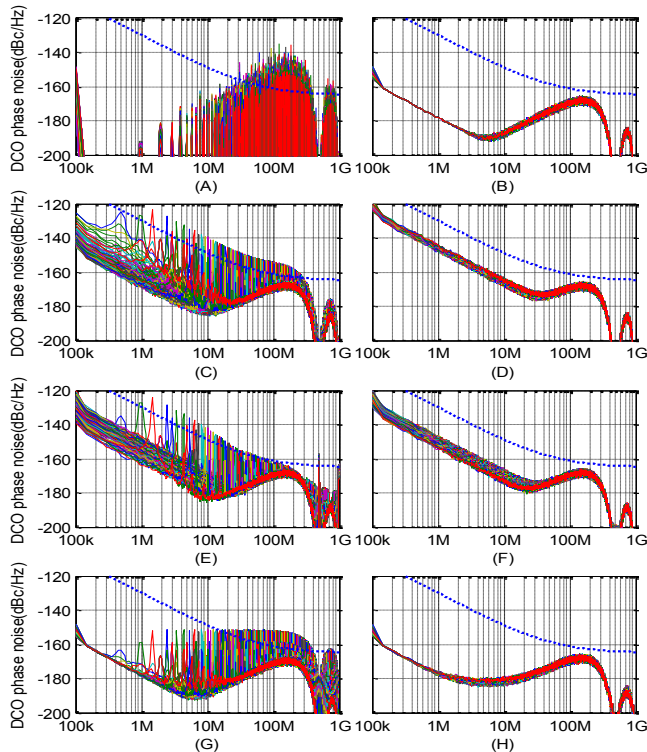


Fig. 9. DCO phase noise without (A, C, E, G) and with (B, D, F, H) the proposed spur mitigation schemes. Spurious behavior in the presence of idle tones (A, B), $\pm 10\%$ varactor mismatch (C, D), 150 ps rise/fall time mismatch (E, F) and 400 ps delay mismatch (G, H).

clock rate of the $\Sigma\Delta$ modulator $f_{\Sigma\Delta}$ is 1/4 of the DCO center frequency of 2 GHz.

First, the DCO interface formed by an *ideal* $\Sigma\Delta$ modulator and *ideal* tuning varactors is simulated without (Fig. 9A) and with (Fig. 9B) the LSB dithering scheme of Fig. 6. Clearly, all idle tones of the ideal $\Sigma\Delta$ modulator are completely suppressed with the LSB dithering scheme, which is then enabled for *all* subsequent simulations.

Second, a set of frequency tuning units with the frequency step sizes of $1.1f^T$, $0.9f^T$ and f^T ($\pm 10\%$ mismatch) were modeled and the simulation results without and with the proposed bit rotator are shown in Fig. 9C and Fig. 9D, respectively. The spurious tones resulting from the varactor mismatch are completely dithered out. Similarly, Fig. 9E and Fig. 9F show the cases without and with the proposed spur mitigation schemes in the presence of 150 ps rising and falling delay mismatch in the oscillator varactor tuning bus. When one tuning bit has 400 ps more delay than other bits, more spurs may appear at high frequencies, as shown in Fig. 9G, and are completely removed once the proposed dithering scheme is enabled (Fig. 9H).

Fig. 10 shows a summary of all noise contributors with the proposed randomization schemes. To better illustrate the results, the figure shows the envelope, i.e., maximum spectral noise energy at each frequency point for all possible inputs. The cumulative dithering noise (bold curve) is approximately 12 dB below the intrinsic oscillator noise (dotted curve) and the results match well with the theoretical analysis in the previous sections.

V. CONCLUSIONS

A spur-free frequency tuning scheme for a digitally controlled oscillator (DCO) has been proposed in this paper.

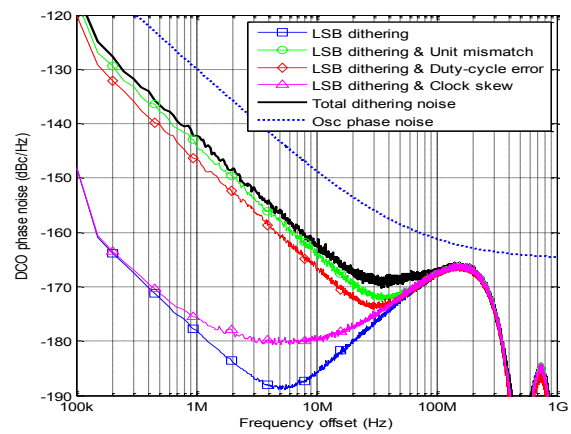


Fig. 10. DCO phase noise contributors with the proposed mitigation schemes in the presence of idle tones, $\pm 10\%$ varactor mismatch, 150 ps rise/fall time mismatch and 400 ps delay mismatch.

The dithering scheme augments the standard $\Sigma\Delta$ MASH modulator with a low-complexity but effective structural modifications. This results in elimination of observable spurs at the DCO output caused by the fine frequency control $\Sigma\Delta$ dithering even with an arbitrary constant frequency tuning code in the presence of physical and interface artifacts, such as varactor mismatch, as well as delay and transition timing mismatches. The feasibility of the proposed scheme has been verified through analysis and behavioral simulations.

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